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Radiation studies performed on the High Luminosity ATLAS TileCal link Daughterboard

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Abstract: The new electronics of the ATLAS Tile Calorimeter for the HL-LHC interfaces the on-detector and off-detector electronics by means of a Daughterboard. The Daughterboard is positioned on-detector featuring commercial SFPs+, CERN GBTx ASICs, ProASIC FPGAs and Kintex Ultrascale FPGAs. The design minimizes single points of failure and mitigates radiation damage by means of a double redundant scheme, Triple Mode Redundancy, Xilinx Soft Error Mitigation IP, CRC/FEC for link data transfer, and SEL protection circuitry. We present an updated summary of the TID, NIEL and SEE qualification tests, and performance studies of the Daughterboard revision 6 design.

Keywords: Radiation damage to electronic components, Front-end electronics for detector readout.

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1 Introduction

The Tile Calorimeter (TileCal) is a sampling calorimeter of the ATLAS Experiment [1]. TileCal is in charge of identifying and measuring hadronic jets and transverse missing energy. As depicted in Figure 1, the electronics for the read-out of all the TileCal are divided in four cylindrical barrels, each composed of 64 wedge-shaped modules where plastic scintillator tiles are used as active material interleaved with steel plates used as absorbers. Light from two sides of a pseudo-projective cell composed by a group of scintillators is collected by wavelength shifting fibers and read out by two photomultiplier tubes (PMTs). To face the higher radiation levels and increased rates of pileup at the High Luminosity Large Hadron Collider (HL-LHC), the TileCal electronics will be upgraded with a new design that will provide continuous digital read-out of all the TileCal with improved timing and better energy resolution.

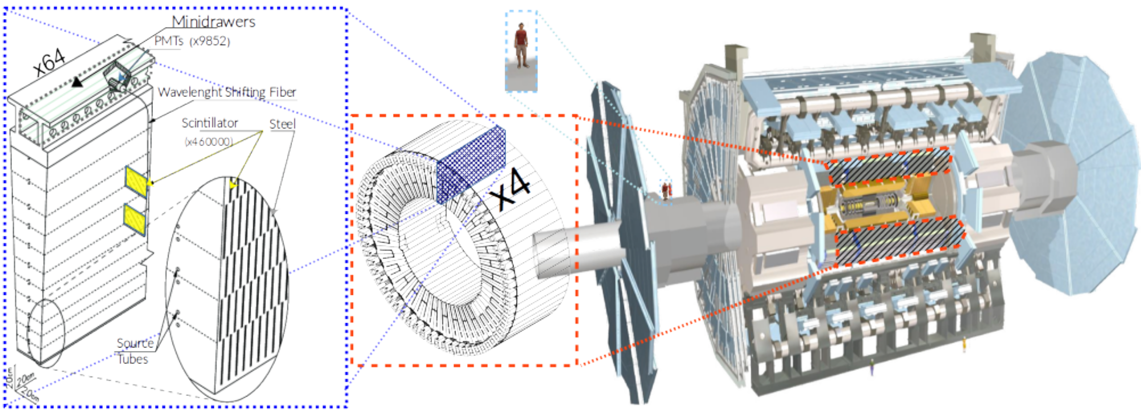

Figure 1. TileCal structure in ATLAS.

2 Tilecal read-out system for the HL-LHC

The off-detector electronics of the upgraded TileCal will provide digitized signals at 40 MHz to the Level 0 trigger system through the Trigger and Data Acquisition interface, while the so-called Front-End Link eXchange (FELIX) system will read-out data stored in pipelines in Tile Preprocessors (TilePPr) at 1 MHz. The TilePPrs will interface the on- and off-detector electronics through multi-Gbps optical links. The power will be monitored and distributed to the on-detector electronics by the Detector Control System that interfaces with the Low Voltage Power Supplies in charge of distributing power to all the digital and analogue electronics, and the High Voltage Power Supplies that will provide high voltage to each PMT.

The on-detector is comprised by 896 Minidrawers (MDs), in which data from all TileCal PMTs will continuously be sampled at 40 MHz. Each MD will host up to twelve channels by means of twelve PMTs that turn light pulses to electric signals to be shaped and conditioned by Front-End Boards. A Mainboard (MB) will continuously sample and digitize two gains of shaped signals of the MD PMTs. A Daughterboard (DB) is in charge of distributing LHC synchronized timing, configuration and control to the front-end, and continuously transmitting the digital data from all the MB channels to the off-detector systems via multi-Gbps optical links.

3 ATLAS TileCal link Daughterboard

The DB is a radiation tolerant board with a redundancy layer of two functionally-equivalent halves, interconnecting the on- and off-detector by means of a 400-pin FMC connector and multi-Gbps optical links, respectively [3]. The DB, currently on its sixth revision (DB6, Figure 2), powers the optical links by means of four SFP+ modules that drive two downlinks and four uplinks. The board information received by the board from the off-detector by means of 4.8 Gbps GBT-FEC downlinks powered by CERN radiation hard GBTx ASICs includes clock signals, time synchronization and configuration commands.

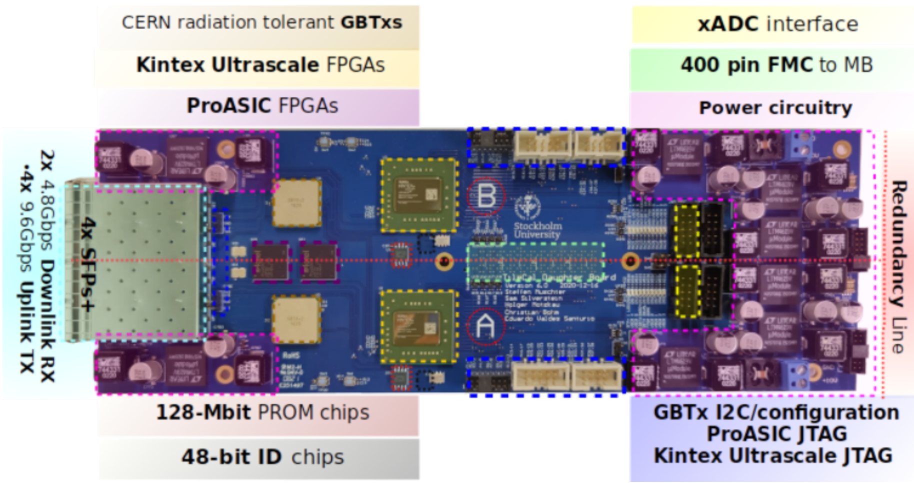

Figure 2. The DB6 layout.

Two 20 nm planar featured Xilinx Kintex Ultrascale (KU) FPGAs distribute clocks and configuration received from the GBTx to the MB, while reading slow integrator data and fast digitized data from two different gains of the PMTs. The xADCs of the KU FPGAs sample temperature, voltage stability and currents. An over-current protection circuit triggers a power cycle in the case of the presence of high currents. Each KU FPGA sends data off-detector by means of two copies of GBT-FEC protected words through two 9.6 Gbps uplinks. The board radiation tolerance is achieved by the redundancy layers in the design, using the Xilinx Soft Error Management (SEM), Triple Mode Redundancy (TMR), GBT-FEC protocol, and GBTx ASICs.

4 Radiation tests on the DB6

Four Daughterboards were separately used to test for radiation tolerance, labelled in this document as DB6-0, DB6-1, DB6-2 and DB6-3.

Two separate tests were performed to cover tolerance to Total Ionizing Dose (TID). The KU FPGA temperatures and all the currents corresponding to all the point of load regulators of the irradiated boards were monitored. Special attention was placed on the total of ten Coretek SFP+, the six KU FPGAs, the six Microsemi ProaSIC3 FPGAs and the IS25LP128F reconfiguration FLASH memories. None of the components were damaged by the deposited TID: DB6-1 and DB6-2 were irradiated with Gamma radiation at ^{60}Co at the CERN CC60 facilities to 220 Gy at a rate of 3.37 Gy/h (Figure 9 of Ref.[4]) and up to 54 Gy at a rate of 0.33 Gy/h (Figure 10 of Ref. [4]), respectively. DB6-1 ran stable during the full test, however it showed an increase in the current measured in the 0.95 V power line of both sides of the board of at around 140 Gy. This was demonstrated to be correlated with the failure of the active components of the fan used to keep the KU FPGAs cool, leading to the increase of temperature on the FPGA. The currents, temperatures and functionalities of the DB6-2 were stable during the full run. DB6-3 was irradiated in the AlC-144 cyclotron proton beam line with 54 MeV protons at the IFJ in Krakow to 381 Gy for half-A FPGA and 400 Gy for half-B FPGA at a rate of 2.88 Gy/h (Figure 3). All the currents and the temperatures were stable during the full run. The higher fluctuations compared with DB6-1 (0 to 140 Gy) and DB6-2 are due to active Single Event Upset (SEU) corrections by the Xilinx SEM and resets applied when an uncorrectable SEU appeared during the run.

Two separate tests were performed for Single Event Latchups (SEL), one as described for the DB6-3 with 54 MeV protons, where the FPGAs were irradiated to a fluence of 2.62×10^{11} p/cm^2. In the second test, two Trenz TE0841 boards (each with the same KU FPGA used in the DB design), a Lattice ICE40LP FPGA and a ProASIC3E A3P31500 FPGA were irradiated with 226 MeV protons to a fluence of 1.11×10^{12} p/cm^2, for which all the currents of the FPGAs were monitored (Figure 6 of Ref. [4]). No SEL were observed in the units under test in either of the tests.

A set of tests was setup to cover the Non-Ionizing Energy Losses (NIEL) tolerance, where a DB (DB6-0) and a set of test boards with non-radiation qualified DB components were irradiated to a fluence of 14×10^{12} (1 MeV equivalent n)/cm^2. The post-radiation tests performed on DB6-0 demonstrated the radiation tolerance of the KU FPGAs up to the exposed fluence. In the case of the ProASIC, some after-effects could be seen where the firmware was not working as expected and re-configuration was not possible. However, the firmware functionality of all devices was recovered after annealing, with re-configuration capability recovered in some of the devices. The same applies

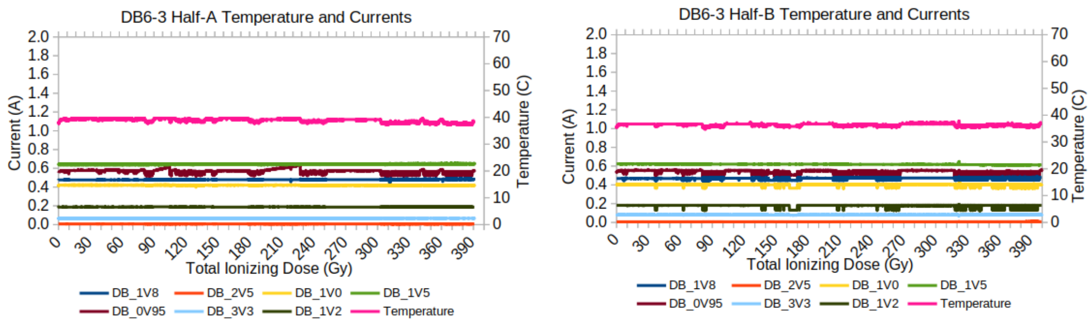

Figure 3. TID test of DB6-3. All board currents and the temperatures of both KU FPGAs were monitored over the whole irradiation period delivered by a 54 MeV proton beam.

to the IS25LP128F FLASH memories, where after annealing some of the devices recovered the re-programing capabilities, with all the contents of the memory not being affected in any of the devices.

Four different variants of 10 Gbps multi-mode SFP+ devices were tested: AVAGO, CORETEK, FS-Europe and Direktronik. The two AVAGO AFBR-709SMZ devices completely failed to pass the irradiation tests done up to 9 $\times 10^{12}$ (1 MeV equivalent n)/cm^2 fluence. One out of nine CORETEK CT-A000NPP-SB1L-D devices tested failed to function after irradiating to 9 $\times 10^{12}$ (1 MeV equivalent n)/cm^2 fluence. In the case of the Direktronik 33-4248 and the FS-Europe SFP-10GSR-85 devices, one of each of the two brands was exposed to a higher fluence of 14×10^{12} (1 MeV equivalent n)/cm^2, and a batch of ten of each of the two brands was exposed to 7 $\times 10^{12}$ (1 MeV equivalent n)/cm^2 fluence. All of the samples irradiated from Direktronik and FS-Europe were fully functional after irradiation.

It was seen that the threshold voltages for the FDFMA3N109 MOSFETs used in the design shifted with the exposure (Figure 4) due to the Gamma component present in the reactor cavity (195 Gy). The asymmetrical profile of the flux in the cavity was used to have a perception of the correlation between the shifts of the thresholds in the MOSFETs and the exposed fluence. Four out of fifteen

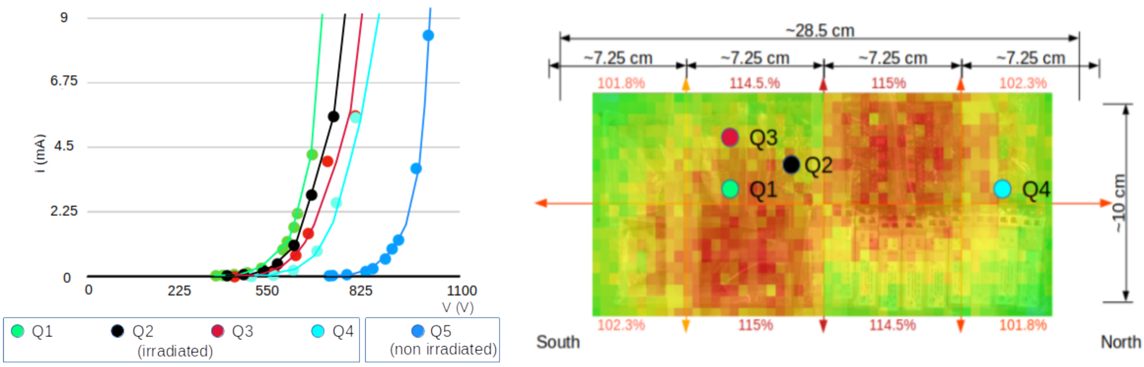

Figure 4. (Left:) Threshold shift depicted in an IV curve of four of the irradiated FDFMA3N109 MOSFETs. (Right:) Position of the irradiated MOSFETs in the flux 2D slice map of the reactor cavity.

QUAD oscillators LTC6902IMS irradiated failed to pass the post irradiation tests, therefore the device needs to be removed from the design. The rest of the components (DAN217T146CT Schottky diodes, 540BAA100M000BBG oscillators and INA333AIDRG instrumentation amplifiers) passed the post radiation tests with no issues.

5 Conclusions

Radiation test campaigns took place to qualify each of the components of the DB6 design to fulfill the HL-LHC requirements. The outcome of the tests have proved that the design is TID and SEL-resistant. The extra protection added to the design for SEL appearances in the form of a over-current circuit has to be removed due to the shift in the threshold of the MOSFETS used in the implementation. Similarly, the QUAD oscillators used to de-couple the phase of the DC-DC converters have to be removed from the design, and extra studies has to be performed to assess the impact of this design modification on the board noise. The removal of the QUAD oscillators and the over-current circuit will make the design sufficiently radiation-hard for the expected NIEL fluence of the full life of the HL-HLC. Future plans include producing extra prototypes that include the aforementioned modifications, performing Single Event Upset tests, and following-up in the tests to finalize the selection of the SFP+ variant to be used with the final board design. After the design is modified accordingly to the test results, prototypes will be produced and tested as part of the road map for the production phase of the project. A total of 930 DB6 will be produced as the contribution of Stockholm University to the HL-LHC era for TileCal.

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